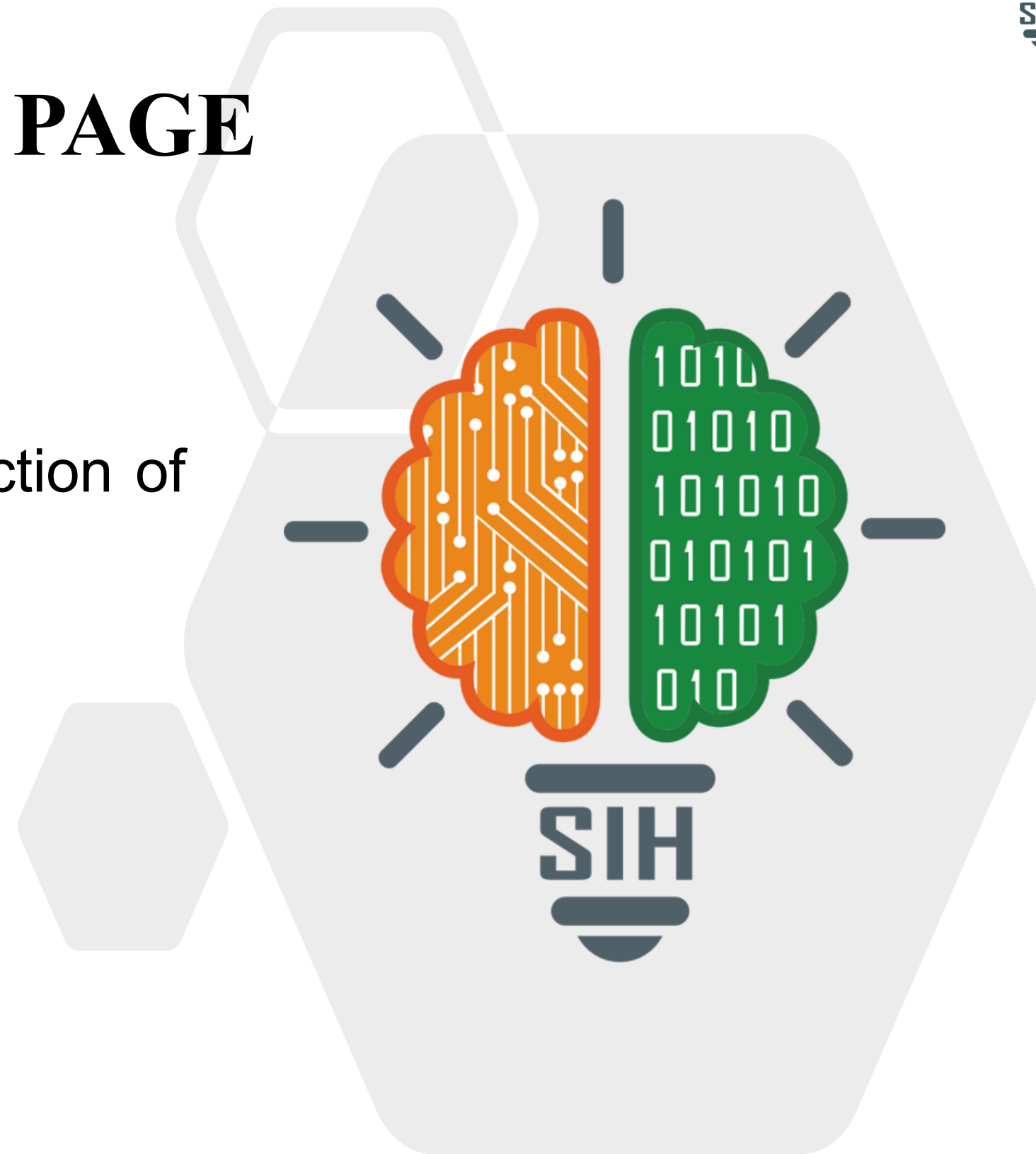


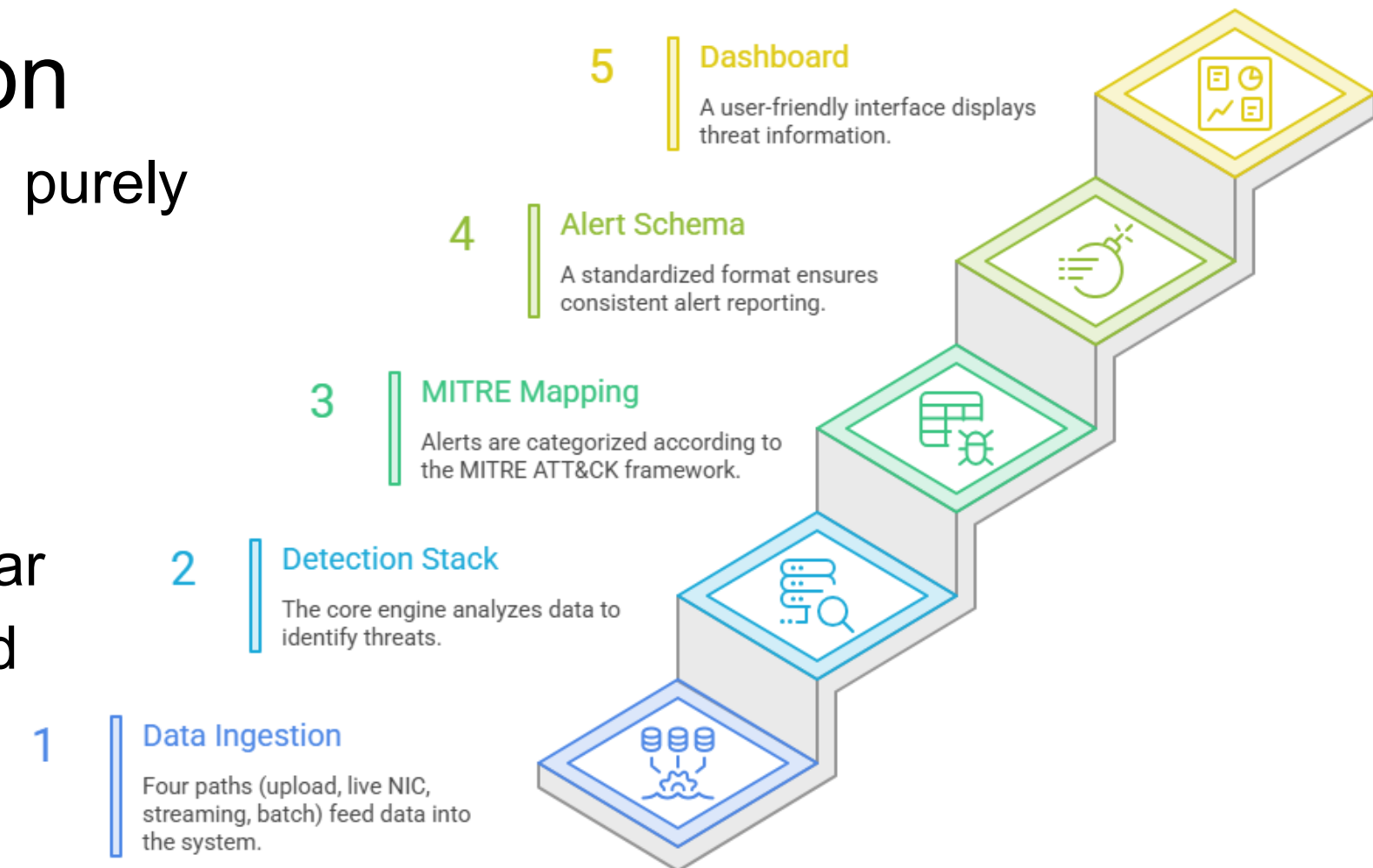
TITLE PAGE

- **Problem Statement ID** – 26145
- **Problem Statement Title-** AI-Based Detection of Cyber Threats in Unidirectional IP Traffic
- **Theme-** Blockchain & Cybersecurity
- **PS Category-** Software
- **Team ID-**
- **Team Name :** TeamXOR



Proposed Solution

- Detailed explanation of the proposed solution
 - A secure, read-only AI pipeline that detects and scores threats purely from network metadata without ever decrypting payloads.
- How it addresses the problem
 - It processes four different data ingest paths through a single, near realtime detection stack that outputs directly to a MITRE mapped dashboard.
- Innovation and uniqueness of the solution
 - It uses specialized mechanisms per threat, hybrid ML to prevent train/serve skew, and JA4 TLS fingerprinting for advanced zero-decryption analysis.

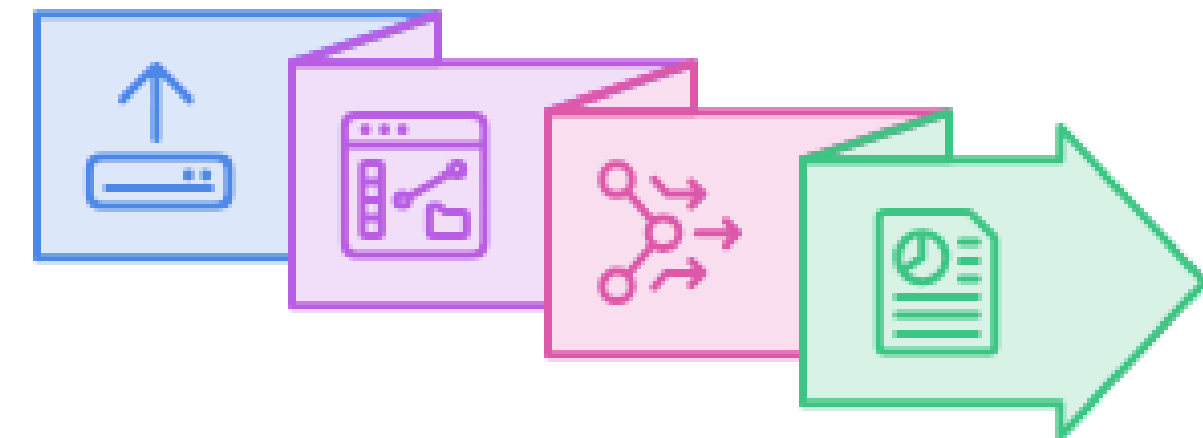


Made with Nankin



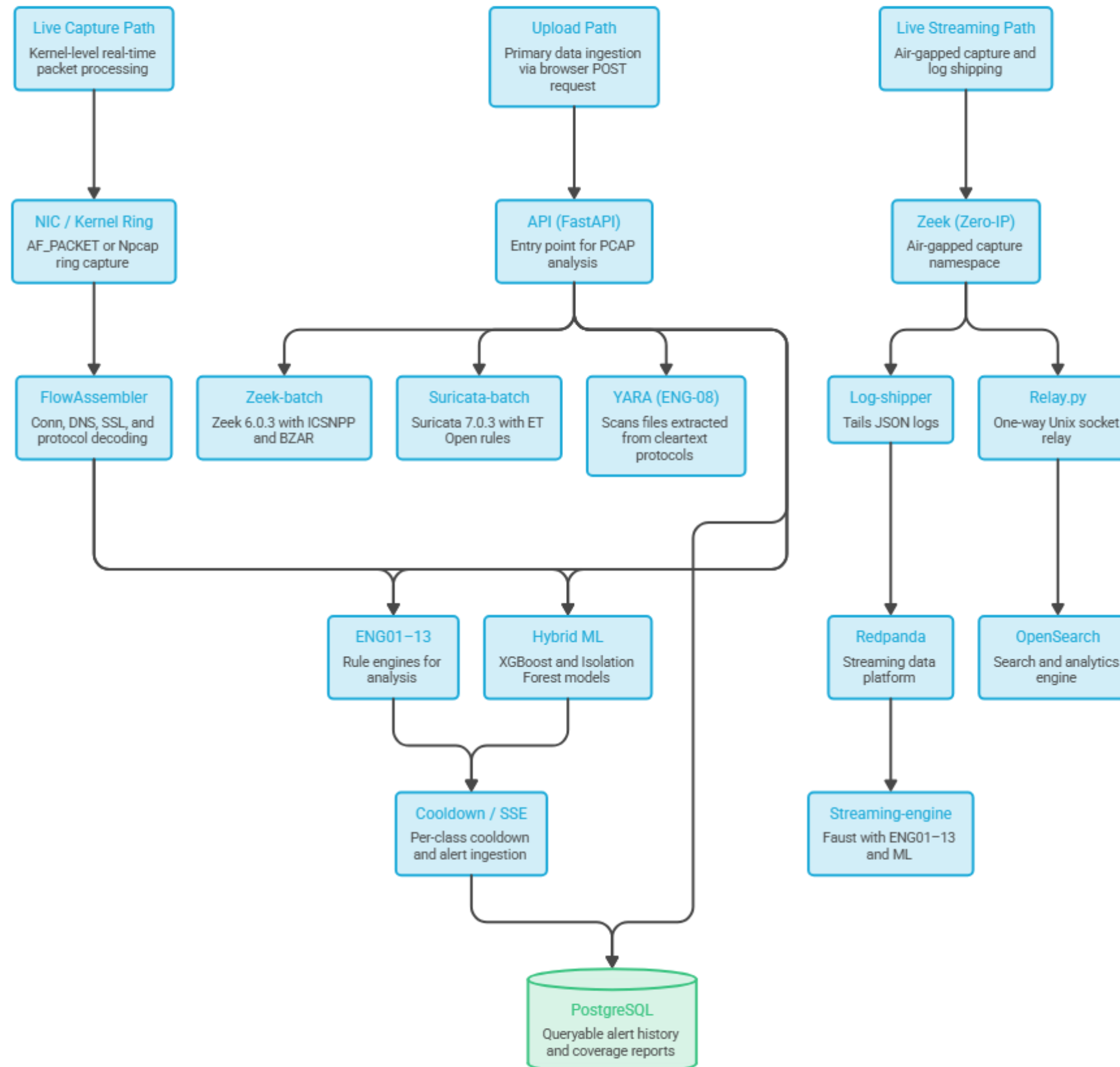
Upload Path
Browser sends PCAP to FastAPI, triggering Zeek 6.0.3, Suricata 7.0.3, YARA, rule engines, hybrid ML, and PostgreSQL storage.

Live Streaming Path
Air-gapped Zeek capture logs via Redpanda to a streaming engine and OpenSearch.



Live Capture Path
Kernel-level capture processes traffic via FlowAssembler, rule engines, and hybrid ML, streaming alerts to PostgreSQL.

Pipeline Coverage Report
Each analysis returns a report to verify the effectiveness of the tools used.



Feasibility

- A fully tested, CPU-only 12-service prototype successfully meets all architectural constraints with high-accuracy models and verified live capture capabilities.

Challenges → strategies

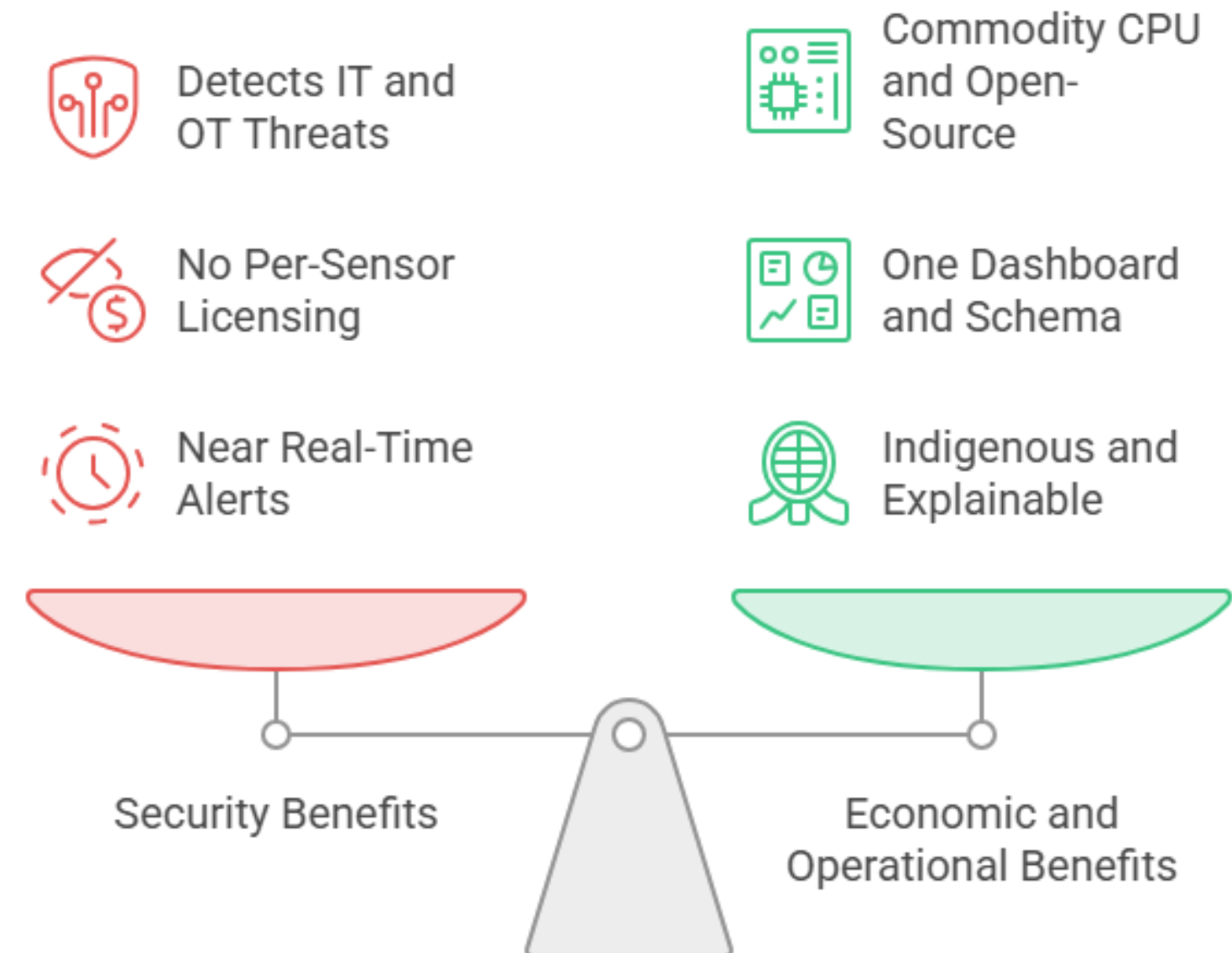
- Technical hurdles like spoofed attacks, link overruns, and data limitations are actively mitigated using entropy tracking, kernel-level optimizations, hybrid model balancing, and targeted retraining pipelines.

Audience:

- critical-infrastructure operators, NTRO / national CERT, SOC & OT analysts.
- Closes the blind spot — detection where you cannot probe back; near real-time, fully auditable alerts.

Benefits:

- Security: IT + OT threats (Modbus/DNP3/EtherNet-IP, lateral movement, Kerberoasting, brute force) from one passive tap.
- Economic: commodity CPU + open-source; no per-sensor licensing.
- Operational / national: one dashboard & schema across all modes; indigenous, inspectable, explainable capability.



Datasets:

- DGA / DNS — chrmor/DGA_domains_dataset (Netlab 360 OpenData): [Github Link](#)
- Flow / DDoS — StopDDoS/packet-captures: [Github Link](#) · upgrade path: CICIDS2017 (UNB CIC): [University of New Brunswick](#)
- OT / Modbus — ITI/ICS-Security-Tools: [Github Link](#) · upgrade path: CIC Modbus 2023 (UNB): [University of New Brunswick](#)
- TLS / JA4 — no public labelled set; planned self-generation.

Tools & frameworks:

- Zeek: [Zeek.org](#) · ICSNPP (CISA): [Github Link](#)
- Suricata: [Suricata.io](#) · Emerging Threats Open: [rules.emergingthreats.net](#) · YARA: [VirusTotal.Github.io](#)
- MITRE BZAR: [Github Link](#) · JA4 spec/intel (FoxIO): [Github Link](#)
- ONNX Runtime: [onnxruntime.ai](#) · XGBoost: [xgboost.ai](#) · scikit-learn: [scikit-learn.org](#)

Standards:

- MITRE ATT&CK: [Attack.mitre.org](#) · ATT&CK for ICS: [Attack.mitre.org/matrices/ics/](#)
- NIST SP 800-82 Rev.3 (OT Security): [Doi.org/10.6028/NIST.SP.800-82r3](#) ·
- IEEE 1815 (DNP3): <https://standards.ieee.org/ieee/1815/5414/>
- Modbus spec: [Modbus.org/specs.php](#)